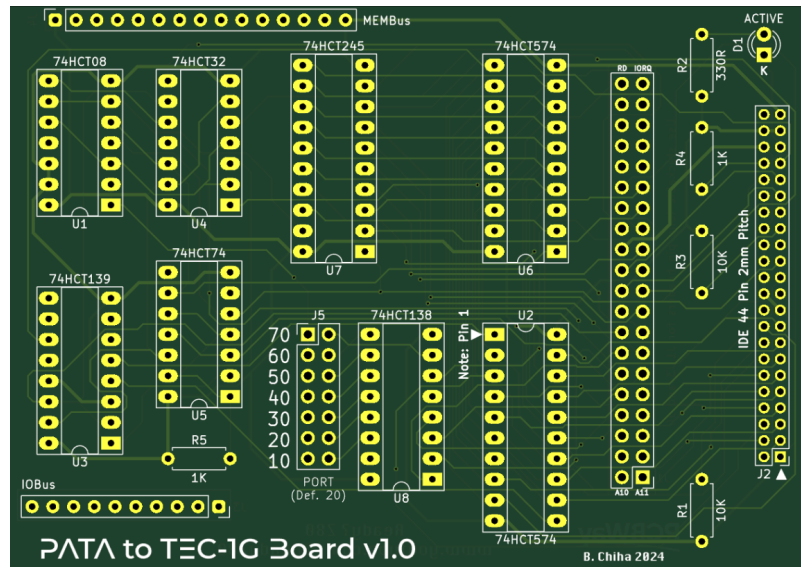


PATA Drive Add-On Board for the TEC-1G

The PATA Drive board allows the TEC-1G to Read files stored on a 44-pin 2.5" IDE HDD or Compact Flash Card. These files can be loaded onto the TEC and executed with some software. This software will be incorporated within Mon3 v1.6 or can be loaded as a stand-alone program. The software can read any disk formatted with the FAT32 file system. At this stage, it will only read the disk. Writing to the disk from the TEC isn't implemented as such.

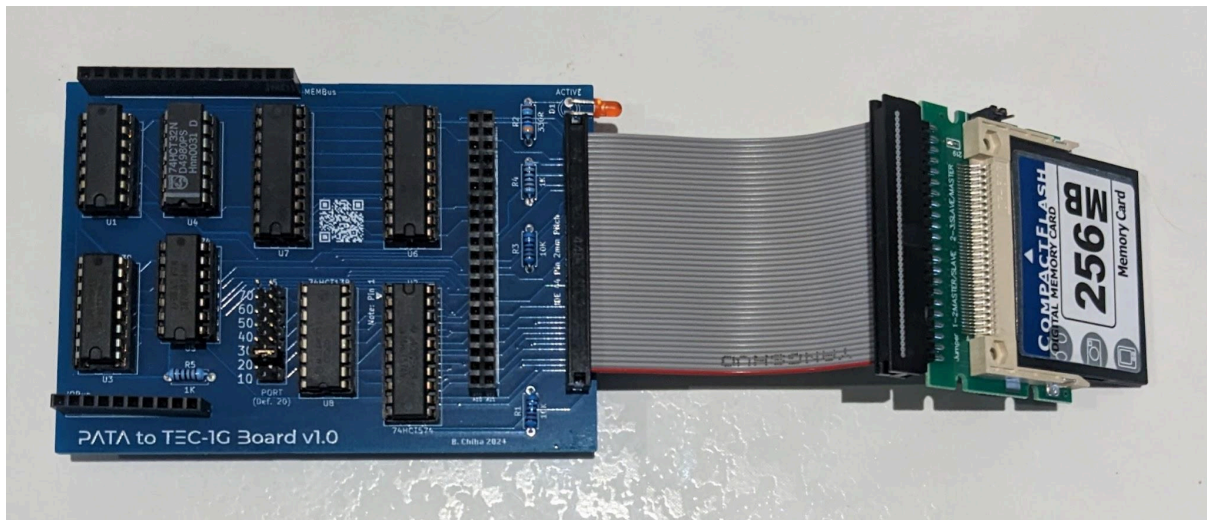
Parts List

Designator	Description
R1,R3	10k Ω Resistor
R2	330 Ω Resistor
R4,R5	1k Ω Resistor
D1	3mm Red LED
U1	74HCT08 Quad AND gate
U2,U6	74HCT574 Octal D Flip Flop latch
U3	74HCT139 Dual 2-Bit decoder
U4	74HCT32 Quad OR gate
U5	74HCT74 Dual D Flip Flop
U7	74HCT245 Octal Bus Transceiver
U8	74HCT138 3-Bit decoder
J1	2x20 long pin stackable header
J2	2x22 header 2mm pitch (IDE)
J3	1x10 long pin stackable header
J4	1x15 long pin stackable header
J5	2x7 header 2.54mm pitch
Socket	3 x 14 pin IC Socket
Socket	2 x 16 pin IC Socket
Socket	3 x 20 pin IC Socket
Block	Header Shortening Block
Cable	2mm 44 pin IDC Cable
Module	IDE to Compact Flash Adaptor
Part	Compact Flash to USB reader
Memory	Pre-Installed Compact Flash card



Construction

1. Place in the 5 resistors. Use a multimeter to ensure correct values are in the right spot
2. Insert the LED, long pin near ACTIVE text. Bend it to the right over the board for better viewing
3. Now insert the 8 IC Sockets, All have pin 1 downwards except **U2** which is upwards.
4. J2 is next, this is probably the hardest as the 2mm pitch is small for soldering. Use a small soldering tip and a little solder. Use a magnifying glass to check for any pin bridges.
5. Install the J5 header next, place the Header Shortening Block horizontally on Port 20
6. Next solder the stackable headers, J1, J3 and J4
7. Insert one end of the IDC cable into J2 and the other into the IDE Adaptor. For neatness, place the red line of cable on pin1 (downwards)
8. Slide the pre-installed CF card into the adaptor.
9. Insert the 8 ICs in their correct position.



Installation

The PATA board is designed to slot into the TEC deck. Slowly place the board down on all 3 connectors. A quick way to see if the board is responding is by typing **DB 27 F7** into the TEC. Run the code and on the LCD Screen, the Register A should be **50**. If it returns **FF** then check the port shortening block is on port 20. If it's **5x**, then it might not sit correctly on the TEC. Move the board slightly up or down on the TEC connectors. **I'm not sure if it's the header contacts but this is my biggest problem with it not being recognised.** Jiggling it a bit and turning the TEC back on eventually fixes this issue. Also, any time the CF card is inserted or the board is moved, ensure the TEC is switched off first.

Software

Software is needed to communicate with the board. The code should be placed on a ROM for easy reuse when powered down. Mon3 v1.6 includes the code as part of the Main Menu or by pressing Fn-F while in data entry mode. If you can burn EPROMS, then I will send you Mon3 v1.6 for you to burn, otherwise, use the pata_fat32.z80 code in the GitHub repository. This code can be compiled on <https://asm80.com>. Ensure you set an **ORG \$2000** to place the code at address 2000. Then load the HEX file to the TEC.



Running Programs

If all is working, when the software is run, the LCD will catalogue the files on the disk. Select the file by using Plus/Minus and press GO to load. If the file ends with HEX then the file is automatically converted to binary and loaded at the correct location in RAM. Any other file will ask for a destination address. The CF card is pre-installed with loads of software for the TEC. See the software listing for more information.

You can also copy your code onto the CF card by using the USB to CF adaptor on your PC or Mac. Simply slide out the CF card from the TEC, and insert it into the CF reader. Your computer will recognise the disk. Only use up to 8 characters for file names otherwise they will be shortened.

Please contact me for any issues or assistance, at brian.chiha@gmail.com

Project Repository: https://github.com/bchiha/Ready-Z80/tree/main/26-PATA_FAT32

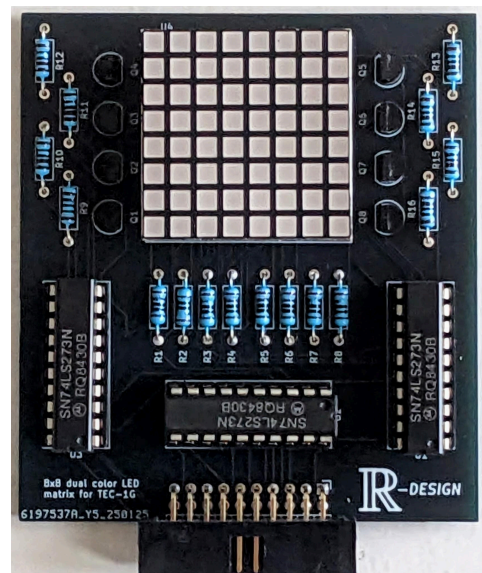
8x8 Matrix Add-On Board

The CF card files contain programs requiring add-on boards for the TEC-1G. I have included a newly designed 8x8 matrix board in this kit. This kit uses an 8x8 Red/Green LED module. It connects to the 20-pin TEC Expander Socket (Top Right). Port 05 is the Row Select, Port 06 is Red, and Port F8 is Green. You are required to source the components for this board yourself. The LCD module can be purchased at Adafruit or Core Electronics.

Parts List

Designator	Description
R1-R8	8 x 220Ω Resistor
R9-R16	8 x 1kΩ Resistor
Q1-Q8	8 x BC547 Transistor
U1-U3	3 x 74LS273 Quad Latch
U4	BL-M12A883DUG 8x8 LED MATRIX
J1	2x20 Right Angle IDC header

Reinoud de Lange designed this board. He has provided it for free to the community.

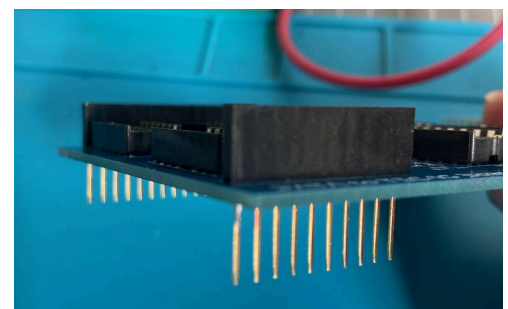
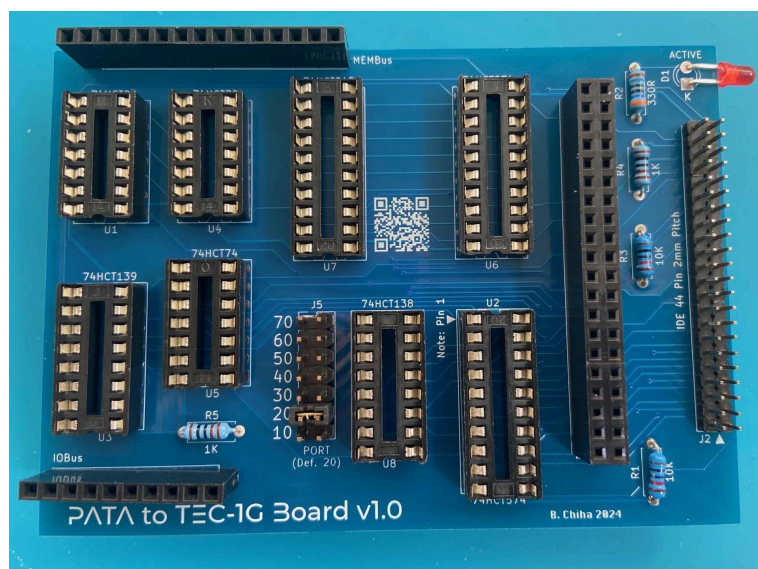


Other Add-Ons

Some further programs use the Graphical LCD or GLCD Board and the Full Matrix Keyboard. Information on these boards can be found at <https://github.com/markjelic/tec-1g>

Finally, here is a video that explains how this project was put together and how to use the software. Look at the Ready? Z80 YouTube channel for the PATA project and much more!

<https://www.youtube.com/@ReadyZ80>



Project Repository: https://github.com/bchiha/Ready-Z80/tree/main/26-PATA_FAT32

